THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINSTRATION AND LOCAL GOVERNMENT

KASULU TOWN COUNCIL

FORM FOUR PRE-MOCK EXAMINATION

CODE: 032 **CHEMISTRY 1**

Time: 3:00 Hours **Thursday, 22.05.2025**

INSTRUCTIONS

1. This paper consists of sections A, B and C with a total of eleven (11) questions.
2. Answer all questions in Sections A and B and two questions from section C.
3. Section A carries fifteen (15) marks, section B carries fifty-five (55) **marks** and section C carries thirty (30) **marks**.
4. Non programmable calculator may be used.
5. Communication devices and any unauthorized materials are not allowed in the examination room.
6. The following constant may be used:

Atomic masses; Cl = 35.5, C = 12, H = 1, Ca = 40, O = 16

Avogadro's number = 6.02×10^{23}

GMV at s.t.p = $22.4dm^3$

1Faraday = 96,500Coulomb

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SECTION A (**15 marks)**

(**Answer all questions in this section)**

1. For each of the items (i)- (x) choose the correct answer from the given alternatives and write its letter in your answer sheet provided:

i) Why do the ships often have blocks of magnesium attached to their hull?

A. To improve appearance of the hull

B. To make the hull stronger

C. To give sacrificial protection to the hull

D. To weigh down the ship in the water

E. To know the total mass of the hull

ii) Diamond is used for cutting glass because:

A. It cannot be spoiled by glass B. It does not react with glass

C. It is very hard D. It does not conduct electricity

E. It is used as ornament.

iii) Colored substances can be separated through the process called:

A. Filtration B. Chromatography E. Deposition.

C. Distillation D. Sublimation

iv) A solution of pH 1.5 is best described as;

A. Weak acid. B. Strong base. E. Weak and strong base exclusively

C. Weak base D. Strong acid

v) Which of following groups of substances represents Flammable liquid;

A. Petrol, Pesticide and Hydrogen. B. Petrol, Sulphuric acid and methylated spirit

C. Methylated spirit, petrol and kerosene. D. Kerosene, diesel and hot water.

E. Petrol, Diesel and hot carbon.

vi) Element M of group I element combines with element X of group VI, the q formula of the compound is:

A. M_2X. B. X_2M_6. C. MX_6. D. MX_2 E. M_6X_2

vii) The number of molecules in 16.8Litre of Chlorine gas at S.T.P is ;

A. 4.516×10^{23} B. 7.5×10^{22}. C. 6.022×10^{23}. D. 9.033×10^{23} E. 4.123×10^{23}

viii) The use of electricity to decompose molten NaCl into its compound element is examples of:

A. Electrolysis. B. Galvanization. C. Hydrolysis.

D. Electroplating. E. Tin planting.

ix) An organic compound has an empirical formula of CH, relative mass of the compound is 28. The molecular formula of the compound is:

A. C_2H_2. B. C_2H_4 C. C_2H_6. D. C_3H_6. E. C_3H_4

x) The total number of protons and neutrons in the nucleus Of an atom is called.....?

A. Valency number. B. Atomic number . C. Molecular number

D. Mass number. E. Electron number.

2. Match each of the molecular formula of the organic compounds in column A with the family names in column B.

COLUMN A	COLUMN B
(i) C_5H_{12}	A. Alkyl
(ii) $C_7H_{14}O_2$	B. Alkaline
(iii) C_4H_6	C. Alkyne
(iv) $C_{17}H_{36}O$	D. Alkane
(V) C_9H_{19}	E. Alcohol
	F. Aldehyde
	G. Carboxylic acid

SECTION B. (55 **marks**)

(**Answer all questions in this section)**

3. The preparation of ammonia in the laboratory is done by heating a mixture of ammonium chloride and sodium Hydroxide;

(a) Write the balanced equation for the above reaction.

(b) Using chemical equation, explain how ammonia reacts with;

(i) Hydrogen Chloride gas

(ii) Heated copper Oxide.

(C) State how ammonia can be used to our daily life, State two (2) points.

4. T and K element in the periodic table the atomic number of T is 16 and that of K is 19.

(a) i. Is the element T a metal or non-metal?

ii. In which group and period does element T appear?

iii. Write a molecular formula of a compound formed between T and K.

(b) i. Which particles are atoms of the same element in the following given particles?

$^{40}_{18}A$, $^{38}_{20}B$, $^{38}_{18}C$, and $^{40}_{19}D$

ii. Give the electronic configuration of sodium and neon. If the atomic number of sodium is 11 and that of neon is 10. And state why can't neon react with sodium?

5. (a) State Faraday's law of electrolysis.

(b) An element X has a relative atomic mass of 88, when a current of 0.5amperes was passed through fused chloride of X for 32 minutes and 10 seconds, 0.44g of X were deposited at the cathode.

i. Calculate the number of Faraday needed to liberate one mole of X.

ii. Write the formula of X ion.

iii. Write the formula of Hydroxide of X.

6. Form four students results from Kileya secondary school perform an experiment to know how salts behave on heating, they used limestone as a sample. when heated, the limestone decomposes into compound X and Carbon dioxide gas.

(i) Identify compound X

(ii) Write down a balanced chemical equation for the above reaction.

(iii) If the amount of limestone heated was 150g. Calculate the mass of the compound X formed.

7. (a) Why is important to replace fossil fuels as an energy source?

(b) Briefly explain four (4) alternatives source of energy that can be developed to replace the fossil fuels.

8. (a) Why carbon form many compounds than any other element? Give three, (3) reasons.

(b) A molar solution of 25$cm^3$of sodium hydroxide (NaOH) reacted completely with 12.50cm^3 of Sulphuric acid. Calculate the concentration of Sulphuric acid in g/dm^3

SECTION C (30 marks)

(Answer only two (2) questions from this section)

9. A form four students at Kitonga secondary school had a task to mop the laboratory chamber, accidentally bees attacked and bite one of them on her hands. The technician advised to apply ashes powder to the bite surface. Explain five (5) importance of the process taking place on the skin surface to our daily life after applying the ashes powder.

10. How is organic manure important to soil quality? Briefly explain by giving five (5) points.

11. There is a various environment plans to be used in controlling terrestrial pollution. Advice accordingly the effective plans to be taken in order to achieve such activities. (Five points (5)).

KASULU TOWN COUNCIL

MARKING SCHEME PRE-MOCK EXAMINATION

CHEMISTRY 1 MAY 2025

1@ = 10marks

1.

S/N	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answer	C	C	B	D	C	A	A	C	A	D

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2.

S/N	i	ii	iii	iv	v
Answer	D	G	C	E	A

1@ = 05marks

3 (a) $2NH_4Cl_{(s)} + Ca(OH)_{2(s)} \xrightarrow{Heat} CaCl_{2(aq)} + 2H_2O_{(l)} + 2NH_{3(g)}$

03 MARKS

(b) (i) Ammonia react with Hydrogen chloride gas to form white denser fume of ammonium chloride.

$NH_{3(g)} + HCl_{(g)} \longrightarrow NH_4Cl_{(s)}$ 02 marks

(ii) Ammonia oxidize Copper (II) oxide to nitrogen water and reduced copper.

$CuO_{(s)} + NH_{3(g)} \longrightarrow 3Cu_{(s)} + 3H_2O_{(l)} + N_{2(g)}$

02 MARKS

(c) Uses of ammonia to our daily life.

- Used to manufacture fertilizer.
- Used to manufacture nitric acid.
- Used in refrigeration
- Used to manufacture various organic substances. eg. plastic.

Any two points
1@ = 02MARKS

4. (a) (i) Element T is non metal. 01MARKS

(ii) Element T appear Group VI and Period 3. 02MARKS

(iii) Molecular formular formed when T compain with element K is K_2T. 02MARKS

(b) (i) Particle $^{40}_{18}A$ and Particle $^{38}_{18}C$

(ii) Electronic configuration of Sodium = 2:8:1. 1.5MARKS and Electronic configuration of Neon is = 2:8. 1.5MARKS

Neon and Sodium can not react because Neon do not have charge means has stable element. 01MARK

5. (a) Faraday First law of electrolysis state that "The mass deposited or liberated of a substance during electrolysis is directly proportional to the quantity of electricity passed through through the electrolyte" and. 1.5MARK

Faraday second law of electrolysis state that "when the same quantity of electricity is passed through solutions of different electrolyte, the mass of the substance liberated or deposited at the electrodes are directly proportional to the chemical equivalents of the substances". 1.5MARKS

5 (b) (i) DATA GIVEN

Atomic mass = 88g/mol

Current (I) = 0·5A 01MARK

Time (t) = 32 Minutes and 10second = 1930second

Mass (M) = 0·44g.

Number of Faraday = Required.

from First faradays

$$M = \frac{ArIt}{V \cdot F}$$

Where M - Mass deposited

Ar - Atomic mass

I - Current

t - Time

V - Valence

F - Faraday constant.

$$\frac{0{\cdot}44g}{1} = \frac{88 \times 0{\cdot}5 \times 1930}{96{,}500 \times V}$$

$$0{\cdot}44 \times 96500V = 88 \times 0{\cdot}5 \times 1930$$

$$\frac{42{,}460V}{42460} = \frac{84920}{42{,}460}$$

$$V = 2 \cdot$$ 02 MARKS

1F = 1Mol = 1V

So. 2 Faraday are required to liberate one mole of X.

(i) Number of Faraday = 2 Faraday 01MARK

(ii) Formular of X ion = X^{2+} 01MARK

(iii) Formula of Hydroxide of X = X_2OH 01MARK

6 (i) Compound X is Calcium Oxide (CaO) 1.5 MARKS

(ii) Chemical equation.

$$CaCO_3(s) \xrightarrow{\text{strong heat}} CaO(s) + CO_2(g)$$ 02 MARKS

(iii) DATA GIVEN

Mass of $CaCO_3 = 150g$ 0.5 MARK

Molar mass of $CaCO_3 = 100g/mol$

Mass of $CaO(X)$ = Required.

from

$$\text{Mole(n)} = \frac{\text{mass}}{\text{molar mass}}$$

$$n \text{ of } CaCO_3 = \frac{150g}{100g/mol}$$

$$n \text{ of } CaCO_3 = 1.5 mol.$$ 1.5 MARKS

From balanced chemical equation.

$$CaCO_3 \xrightarrow{\text{Heat}} CaO + CO_2$$ 01 MARK

1 mole of $CaCO_3 \equiv$ 1 mole of CaO

1.5 mole of $CaCO_3 \equiv X$

$X = 1.5$ Mole of CaO. 01 MARK

from

Mass of (CaO) = Mole × molar mass

mass of $CaO = 1.5 \times 56 g/mol$

mass of $CaO = 84g$

1.5 MARKS

∴ The mass of compound X $(CaO) = 84g$.

7. (a) Reasons for the important of replacing fossil fuels as source of energy. Any four (4) point 1@ = 04 MARKS

- Because other source of energy is environmental friendly
- Because other source of energy is cheap, effective and reliable.
- Because other source of energy is easy to transport.
- Because other source of energy produce high amount of heat.
- Because other source of energy is affordable and reliable.

(b) Four alternatives sources of energy.

Any three point explanation 1:5@ = 06 MARKS

- Solar energy
- Biomass energy
- Biogas energy
- Wind energy
- Water power energy.

8. (a) Carbon forms many compound than any other element because:-

- It has ability to form single and multiple bond.
- It has ability to form long compound.
- It has ability to form strong covalent bond
-

Any three point

1@ = 03 MARKS.

8 (b) DATA GIVEN

Volume of NaOH (V_b) = $25cm^3$

Volume of H_2SO_4 (V_a) = $12.50cm^3$

Molarity of base (M_b) = 1M (Molar solution)

Molarity of acid (M_a) = Required (x) ½ MARKS

Concentration of acid = Required (x)

From balanced chemical equation

$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$ ½ MARK

Number of mole of acid (n_a) = 1

Number of mole of base (n_b) = 2.

from the formula

$$\frac{M_a V_a}{M_b V_b} = \frac{n_a}{n_b}$$ 01 MARK.

Where

M_a – molarity of acid

M_b – molarity of base

V_a – volume of acid.

V_b – volume of base

n_a – number of mole of acid.

n_b – number of mole of base

$$\frac{M_a \times 12.50}{1 \times 25} = \frac{1}{2}.$$

$$\frac{25M_a}{25} = \frac{25}{25}$$

$M_a = 1M$ 02 MARKS

From again

Concentration = Molarity × molar mass

= 1M × 98g/mol.

= $98g/dm^3$

∴ The concentration of acid is $98g/dm^3$.

02 MARKS

9. Importance of neutralization reaction.

Essay
Introduction = 2.5
Point (2@) = 10marks
Conclusion = 2.5marks
Total = 15marks

- Treating insects stings and bites
- Important in soil treatment.
- Important in manufacturing of fertilizer.
- Important in treating factory wastes
- Important in Relieving indigestion.
- Important in neutralising accidental spills.

10 Importance of organic manures

- Manure do not change the pH of the soil.
- Manure give soil a characteristics of dark colouration
- It improve structure of the soil.
- It improve soil fertility by releasing nutrient into the soil.
- Manure are not readily leached out.
- Manure provide food and shelter for soil microorganism, which is responsible for decomposition of organic matter.

Essay format
Introduction = 2.5marks
Main body = (2@) = 10marks
Conclusion = 2.5marks

11- Environmental plans to control terrestrial pollution.

- Use of designated dumping sites.
- Use of biodegradable plastics.
- Re-use and recycling of waste materials.
- Creating awareness
- Formulating and enacting policies and laws.
- Stopping deforestation.

essay format
Introduction 2.5 marks
Main body (2@) = 10marks
Conclusion = 2.5marks
Total = 15marks.